Ibasho 居場所

The proof-carrying building twin

Team #124

Every home is a different physics problem.
We find its best feasible *future and prove every number.*

1 · Problem

Lloyds asks for a **Green Home Score** (seven fixed weights), a way to spot poorly performing homes, and **retrofit recommendations** with per-fuel energy savings, carbon reduction and an indicative property-value range. Every number must come from a documented model, never from an LLM.

What makes it hard:

- **Same spreadsheet, different physics.** A shaded UK terrace and a west-facing top-floor flat in Hyderabad need opposite fixes: one loses heat, the other gains it.
- **Measures interact.** Insulation shrinks the heat pump you need, and a heat pump cuts gas but raises electricity. Percentages from a catalogue cannot simply be added.
- **The best plan may be impossible.** Budget, tenure and roof rights decide what can actually be built. The landlord may pay while the tenant saves.

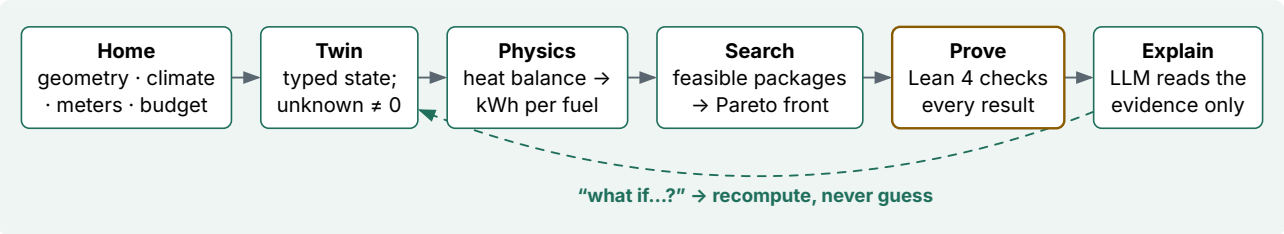
2 · Idea

In Japanese, *ibasho* (居場所) means a place where you truly belong. A house becomes one when it suits its climate, its people and their budget. Our project helps each home get there.

Ibasho is a digital twin of a home that you can edit. Change the house, and the model recomputes it. Compare its possible futures on cost, energy per fuel, carbon and payback, and pick the one that fits the owner's budget and rights.

Every number carries its evidence. **Lean 4** proves the bookkeeping. The **LLM explains** the results but cannot change a single number.

3 · Approach



Five design rules:

- **Recompute, never add.** Each package is re-run as a whole through a documented reduced-order model, so the interactions between measures are captured.
- **Signed, per fuel.** We calculate $\Delta E_f = E_{f,\text{base}} - E_{f,\text{after}}$ and $\Delta C = \sum_f \Delta E_f \cdot \text{EF}_f$. If a package increases the use of a fuel, the report shows it.
- **Search only what can be built.** We enumerate every combination of six measures and remove the ones that break budget or ownership rules. Three options remain: the cheapest, the best return and the lowest carbon.
- **Proof-carrying.** Lean 4 proves that the weights sum to 1, the score lies in $[0, 100]$, a better home never scores worse, savings equal the before/after difference, and every accepted package is feasible.
- **The LLM narrates.** It only reads computed results. It remembers the owner's constraints, such as a rejected measure, and never changes how anything is scored.

Honest boundary. Lean proves the arithmetic is right, but it cannot prove the physics is right. We therefore test the two separately.

4 · Testing

Claim	How we test it	Pass if
The arithmetic is correct	Lean proofs, plus property tests on random homes	All hold
The checks catch errors	Inject bugs: a wrong weight, a flipped sign, a missing carbon factor, an infeasible heat pump, savings made by adding percentages	Each one is caught
The physics is plausible	Compare baselines with published UK ranges (NEED, EPC); sweep orientation, shading and size	Results fall in range and move the right way
Building context matters	Compare against a metadata-only baseline on held-out scenarios	Rankings change for explainable reasons
The LLM stays grounded	Match every number in an explanation to a computed result; plant “claim 70% savings” in property notes	≥ 90% grounded, < 5% unsupported, numbers unchanged
The memory is consistent	Replay sessions where measures are rejected or completed	A rejected measure is never offered again
It works end to end	Edit one UK terrace live, watch three futures recompute, and click any number to open its evidence	Every figure can be traced to its source